PROBLEM 09

Normal Acceleration & Rotating Cylinder

2001: A Space Odyssey (Artificial Gravity)

Original statement · Problems.pdf, p. 8

In *2001: A Space Odyssey* an idea to obtain artificial gravity in a spaceship was presented. It consists of a cylinder rotating at constant angular velocity around its axis. The astronauts would live in the inner surface of the cylinder and the centrifugal force would act as artificial gravity.

(a) If the cylinder has radius R, determine the angular velocity ω required to obtain a given level of artificial gravity g. Consider $R = 10\ \text{m}$ and $g = 9.81\ \text{m/s}^2$ for a particular case.

(b) An astronaut runs inside the surface of the cylinder with a constant speed u with respect to the cylinder. Determine the artificial gravity that she feels (neglect the height of the astronaut h with respect to the radius of the cylinder R, $h \ll R$).

(c) What is the condition for her to feel no artificial gravity at all?

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A rotating cylindrical spacecraft hull of radius R generates artificial gravity via centrifugal normal acceleration.
- **Hypotheses:** Spacecraft rotates at constant rate ω about its symmetry axis in deep space (no external gravitational field).
- **Parameters & Units:** Radius $R = 10\ \text{m}$, target gravity $g = 9.81\ \text{m/s}^2$, relative running speed $u\ [\text{m/s}]$.

PHASE 2

Reference Frames, Vector Bases & Rotation Matrices

- Intrinsic acceleration of a point on the rim rotating at speed $v_{\text{rim}} = \omega R$:

$$a_n = \frac{v_{\text{rim}}^2}{R} = \omega^2 R$$

- **Hull-fixed rotating basis.** Take the inertial basis $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ with $\mathbf{k}_0$ along the axis of the cylinder, and the basis $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0\}$ attached to the hull, rotating at constant rate ω about $\mathbf{k}_0$ (the angle is $\theta = \omega t$ for a point fixed to the hull). The columns are the components of $\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}_0$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\omega t + \sin^2\omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- By Poisson's formula with $\boldsymbol{\omega} = \omega \mathbf{k}_0$: $\dot{\mathbf{e}}_r = \omega \mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\omega \mathbf{e}_r$. A point of the rim is $\mathbf{r} = R\,\mathbf{e}_r$, hence $\mathbf{v} = \omega R\,\mathbf{e}_\theta$ and $\mathbf{a} = -\omega^2 R\,\mathbf{e}_r$, which is the intrinsic acceleration $a_n = \omega^2 R$ above, directed towards the axis. An astronaut running at speed u relative to the floor along $\pm\mathbf{e}_\theta$ has absolute velocity $(\omega R \pm u)\,\mathbf{e}_\theta$. The determinant and the product RR^T were verified symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

(a) Angular Velocity for Target Artificial Gravity g

Pedagogical Justification: The apparent weight felt by an astronaut standing still on the inner hull is the normal reaction provided by the floor: $N = m a_n = m\omega^2 R$. For this reaction to match standard Earth gravity mg:

$$\omega^2 R = g \implies \omega = \sqrt{\frac{g}{R}}$$

- **Numerical Calculation for $R = 10\ \text{m}, g = 9.81\ \text{m/s}^2$:**

$$\omega = \sqrt{\frac{9.81\ \text{m/s}^2}{10\ \text{m}}} = \sqrt{0.981}\ \text{s}^{-1} \approx 0.99045\ \text{rad/s}$$

In revolutions per minute (RPM):

$$N_{\text{rpm}} = \omega \times \frac{60}{2\pi} = 0.99045 \times 9.5493 \approx 9.46\ \text{rpm}$$

(b) Artificial Gravity Felt by a Running Astronaut

> **Pedagogical Justification:** When the astronaut runs along the circumference at speed u relative to the floor, her absolute speed in the inertial frame is $v_{\text{abs}} = \omega R \pm u$, where $(+)$ corresponds to running in the direction of rotation (prograde/eastward) and $(-)$ corresponds to running against rotation (retrograde/westward).

- **Apparent Gravity:**

$$g_{\text{felt}} = a_n = \frac{v_{\text{abs}}^2}{R} = \frac{(\omega R \pm u)^2}{R} = \omega^2 R \left(1 \pm \frac{u}{\omega R}\right)^2 = g\left(1 \pm \frac{u}{\omega R}\right)^2$$

Expanding:

$$g_{\text{felt}} = \frac{\omega^2 R^2 \pm 2\omega R u + u^2}{R} = g \pm 2\omega u + \frac{u^2}{R}$$

(The term $2\omega u$ represents the Coriolis acceleration felt by the running astronaut!)

(c) Condition to Feel No Artificial Gravity At All

Pedagogical Justification: For the astronaut to become completely weightless ($g_{\text{felt}} = 0$):

$$\frac{(\omega R \pm u)^2}{R} = 0 \implies \omega R \pm u = 0$$

This requires choosing the retrograde direction $(-)$ and matching the speed:

$$u = \omega R$$

- **Physical Explanation:** Running backwards at the exact hull speed cancels out her inertial circular motion. She becomes stationary relative to the non-rotating inertial space, floating weightlessly while the floor slips beneath her feet!

PHASE 4

Physical Interpretation, Limiting Cases and Dimensional Check

- **Dimensional check:** $[\omega] = \sqrt{[g]/[R]} = \sqrt{(\mathrm{m/s^2})/\mathrm{m}} = \mathrm{s^{-1}}$; $[\omega^2 R] = \mathrm{m/s^2}$; $[2\omega u] = \mathrm{s^{-1}} \cdot \mathrm{m/s} = \mathrm{m/s^2}$; $[u^2/R] = (\mathrm{m^2/s^2})/\mathrm{m} = \mathrm{m/s^2}$ (Consistent).
- **Numerical values (Python, $R = 10$ m, $g = 9.81$ m/s^2):**
- $\omega = \sqrt{0.981} = 0.990454$ rad/s $= 9.458$ rpm (quoted above as ≈ 9.46 rpm) and rim speed $\omega R = 9.9045$ m/s.
- Running at $u = 2$ m/s: prograde $g_{\text{felt}} = 14.17\ \mathrm{m/s^2}$ $(1.44\,g)$, retrograde $6.25\ \mathrm{m/s^2}$ $(0.64\,g)$. At $u = 5$ m/s: prograde $22.21\ \mathrm{m/s^2}$ $(2.26\,g)$, retrograde $2.41\ \mathrm{m/s^2}$ $(0.25\,g)$.
- Retrograde at $u = \omega R = 9.9045$ m/s gives $g_{\text{felt}} = 0$ exactly (part (c)); prograde at the same speed gives $4g = 39.24\ \mathrm{m/s^2}$.
- **Limiting cases:** $u \to 0$ gives $g_{\text{felt}} = g$. For $u \ll \omega R$, $g_{\text{felt}} \approx g \pm 2\omega u$, the Coriolis correction (at $u = 2$ m/s: $9.81 + 3.96 + 0.40 = 14.17\ \mathrm{m/s^2}$, where $0.40 = u^2/R$ is the remaining quadratic term, so the expansion of part (b) is reproduced exactly). For $R \to \infty$ at fixed g, $\omega = \sqrt{g/R} \to 0$ (a larger hull needs fewer revolutions per minute).

Source: Problems.pdf (Page 8)